AS · Unit 1 · WPH11 Mechanics & Materials

Moments

A force can make something turn as well as move. How strongly it turns depends on how big the force is and how far its line of action is from the pivot, which is why door handles sit far from the hinges and a long spanner undoes a stiff nut. This page covers moments, centre of gravity and the principle of moments: everything you need for beams, planks, ladders and see-saws.

The moment of a force

Try opening a door by pushing right next to the hinge. It barely moves. Push at the handle with the same force and it swings easily. Push on the handle but directly *towards* the hinge, and nothing happens at all. The turning effect of a force, its **moment**, depends on the force and on how far its **line of action** passes from the pivot.

Moment of a force

FORMULA SHEET

$$\text{moment of force} = Fx$$

F: force / N

x: perpendicular distance from the line of action of the force to the axis of rotation / m

moment / N m

DEFINITION

Moment of a force

The moment of a force about a point is the force multiplied by the **perpendicular distance** from that point to the **line of action** of the force.

The **line of action** is the straight line along which the force acts, extended as far as you like in both directions. The distance x is measured at right angles to it. That one word, *perpendicular*, is where most of the marks are. Moments also have a sense of rotation: each one is either **clockwise** or **anticlockwise** about the point you choose.

Forces at an angle

When the force isn't at right angles to the spanner (or door, or beam), there are two equivalent ways to find the moment. Both give the same answer.

Two ways to get the moment of a force at an angle

1. A force F acts at a distance d along a spanner from the pivot, at an angle θ to the spanner.

2. **Method 1: perpendicular distance.** The line of action passes the pivot at a perpendicular distance $x = d\sin\theta$ (right-angled triangle with hypotenuse d):

$$\text{moment} = Fx = F\left(d\sin\theta\right)$$

3. **Method 2: resolve the force.** Split F into a component along the spanner, $F\cos\theta$, and one perpendicular to it, $F\sin\theta$. The component along the spanner points straight through the pivot, so its moment is zero. Only the perpendicular part turns:

$$\text{moment} = \left(F\sin\theta\right)d$$

4. The two are the same product, just grouped differently:

$$\boxed{\text{moment} = Fd\sin\theta}$$

It is biggest when $\theta = 90^\circ$ (push at right angles) and zero when $\theta = 0$ (push along the spanner, through the pivot).

COMMON MISTAKE

Don't multiply the force by the distance to the point where it acts unless the force is at right angles to that distance. A force whose line of action passes **through** the pivot has zero moment about it, however large it is.

WORKED EXAMPLE

A mechanic pulls on the end of a 0.25 m spanner with a force of 80 N at 60° to the spanner. (a) Calculate the moment about the nut. (b) What force at right angles to the spanner, at the same point, would give the same moment?

Show solution

$$\text{(a)}\quad x = 0.25\sin 60^\circ = 0.217\ \text{m}, \qquad \text{moment} = 80 \times 0.217 = 17.3\ \text{N m}$$

$$\text{(b)}\quad F = \frac{17.3}{0.25} = 69.3\ \text{N}$$

Pulling at right angles does the same job with about 11 N less effort.

EXAM TIP

The unit of moment is the newton metre, N m. It has the same base units as the joule (kg m^2 s^{-2}), but a moment is not an energy, so never write a moment in J. Always state the point you are taking moments about, and whether each moment is clockwise or anticlockwise.

The principle of moments

A see-saw balances when the turning effects on each side cancel. That idea, generalised to any object and any point, is the principle of moments.

DEFINITION

Principle of moments

For a body in equilibrium, the sum of the clockwise moments about any point equals the sum of the anticlockwise moments about the same point.

KEY IDEA

The two conditions for equilibrium

An extended body is in equilibrium only if **both** of these are true:

1. The **resultant force is zero**, so it has no linear acceleration (Newton's first law).
2. The **resultant moment about any point is zero**, so it has no rotational acceleration (the principle of moments).

If the resultant force is zero, the total moment comes out the same about every point, so if it balances about one point it balances about all of them. You can choose whichever point makes the algebra easiest.

WORKED EXAMPLE

A child of weight 300 N sits 1.5 m from the pivot of a uniform see-saw, which is pivoted at its centre. (a) Where must a 450 N child sit to balance it? (b) The see-saw weighs 200 N. What is the upward force from the pivot?

Show solution

(a) Take moments about the pivot. The see-saw's own weight acts at the pivot, so it has no moment there:

$$300 \times 1.5 = 450 \times x \quad \Rightarrow \quad x = 1.0 \text{ m}$$

(b) The resultant force is zero too, so the pivot pushes up with the total weight: $300 + 450 + 200 = 950$ N.

EXAM TIP

Take moments about the point where an unknown force acts that you **don't** want yet. That force passes through the point, so it has zero moment and drops out of the equation, leaving one unknown to solve for. Then use "upward forces = downward forces" for the rest.

Centre of gravity

DEFINITION

Centre of gravity

The centre of gravity of a body is the point through which the whole weight of the body can be taken to act.

Every tiny piece of an object has its own weight, and each of those little weights has its own moment about any point you pick. The centre of gravity is the one point where you can put the **whole** weight and get exactly the same total moment. That's why, for moments, you can replace a beam by a single arrow at its centre of gravity.

- For a **uniform** object (the same material and thickness throughout), the centre of gravity is at the geometric centre: the middle of a uniform rod, the centre of a uniform disc.
- For a symmetrical object, it lies on every line of symmetry.
- It doesn't have to be inside the object: the centre of gravity of a ring is in the empty middle, and that of a boomerang is in the air between its arms.

You'll also hear the term **centre of mass** NOT IN SPEC. In a uniform gravitational field, like the one near the Earth's surface, it is the same point.

Where is the centre of gravity of a loaded rod?

1 A rod has a weight W_1 whose centre of gravity is at distance x_1 from one end, and a load W_2 is fixed at distance x_2. We want the single point $\bar{x}$ where the total weight $W_1 + W_2$ gives the same moment.

2 Take moments about the end. The total moment of the separate weights must equal the moment of the total weight acting at $\bar{x}$:

$$W_1 x_1 + W_2 x_2 = (W_1 + W_2)\,\bar{x}$$

3 Rearrange:

$$\boxed{\bar{x} = \frac{W_1 x_1 + W_2 x_2}{W_1 + W_2}}$$

It is a weighted average of the positions. A 1 m rod of weight 4 N with a 1 N load on one end has its centre of gravity at $(4 \times 0.5 + 1 \times 1.0)/5 = 0.60$ m: shifted towards the load, but not all the way.

Finding the centre of gravity by experiment

SUGGESTED EXPERIMENT

Centre of gravity of an irregular rod

The spec suggests this one. Two methods:

1. **Balance it.** Rest the rod on a knife edge (or a triangular prism) and slide it until it balances. The centre of gravity is directly above the knife edge, because only then does the weight have zero moment about it. Read the position off a metre rule.
2. **Use moments.** If the rod is too long or awkward to balance alone, hang a known weight from one end and find the new balance point. Measure the distances from the knife edge and use the principle of moments to calculate where the rod's own centre of gravity is (you need the rod's weight from a balance).

Using a sharp knife edge reduces the uncertainty in where the pivot is. Approach the balance point from both sides and take the mean, and read the scale at eye level to avoid parallax.

WORKED EXAMPLE

A non-uniform rod 1.00 m long has a weight of 4.0 N. A 2.0 N weight is hung from the 0 cm end, and the rod then balances on a knife edge at the 40 cm mark. Where is the rod's centre of gravity?

Show solution

Take moments about the knife edge. The 2.0 N weight is 0.40 m to the left, so it gives an anticlockwise moment. The rod's weight must give an equal clockwise moment, so its centre of gravity is a distance d to the right:

$$2.0 \times 0.40 = 4.0 \times d \quad \Rightarrow \quad d = 0.20 \text{ m}$$

The centre of gravity is at the $40 + 20 = 60$ cm mark.

Irregular flat shapes: the plumb-line method

To find the centre of gravity of a flat, irregular card (a **lamina**), make a few small holes near its edge. Hang it from a pin through one hole so it can swing freely, and hang a **plumb line** (a weight on a thread) from the same pin. When the card stops, draw along the thread. Repeat from a second hole. The centre of gravity is where the lines cross; a line from a third hole is a check.

Why does it work? While the centre of gravity is to one side of the pin, the weight has a moment about the pin that swings the card back. The card can only be at rest when that moment is zero, which means the line of action of the weight passes through the pin: the centre of gravity is **directly below** the pin, on the plumb line.

Free-body diagrams of extended bodies

For a particle, it didn't matter where you drew each force. For an extended body it matters a great deal, because the position of each force sets its moment. The rules:

- Draw the **weight** at the centre of gravity, vertically down.
- Draw each **support force** where the contact is. A support under a beam pushes up. A **smooth** wall or surface can only push at right angles to itself (no friction).
- A rough floor can supply **friction** along its surface as well as a normal reaction.
- Then apply both equilibrium conditions: forces balance **and** moments balance.

WORKED EXAMPLE

A uniform plank 4.0 m long and of weight 200 N rests on supports at each end, A and B. A person of weight 600 N stands 1.0 m from A. Calculate the forces from the two supports.

Show solution

Take moments about A, so the unknown force at A drops out. The plank's weight acts at its centre, 2.0 m from A:

$$R_B \times 4.0 = 200 \times 2.0 + 600 \times 1.0 = 1000 \quad \Rightarrow \quad R_B = 250 \text{ N}$$

Forces balance vertically:

$$R_A = 200 + 600 - 250 = 550 \text{ N}$$

Check with moments about B: $R_A \times 4.0 = 200 \times 2.0 + 600 \times 3.0 = 2200$, so $R_A = 550$ N. The support nearer the person takes more of the load.

WORKED EXAMPLE

The same plank now rests on support A at one end and support B 3.0 m from A, so 1.0 m overhangs. How far from A can the 600 N person walk before the plank tips?

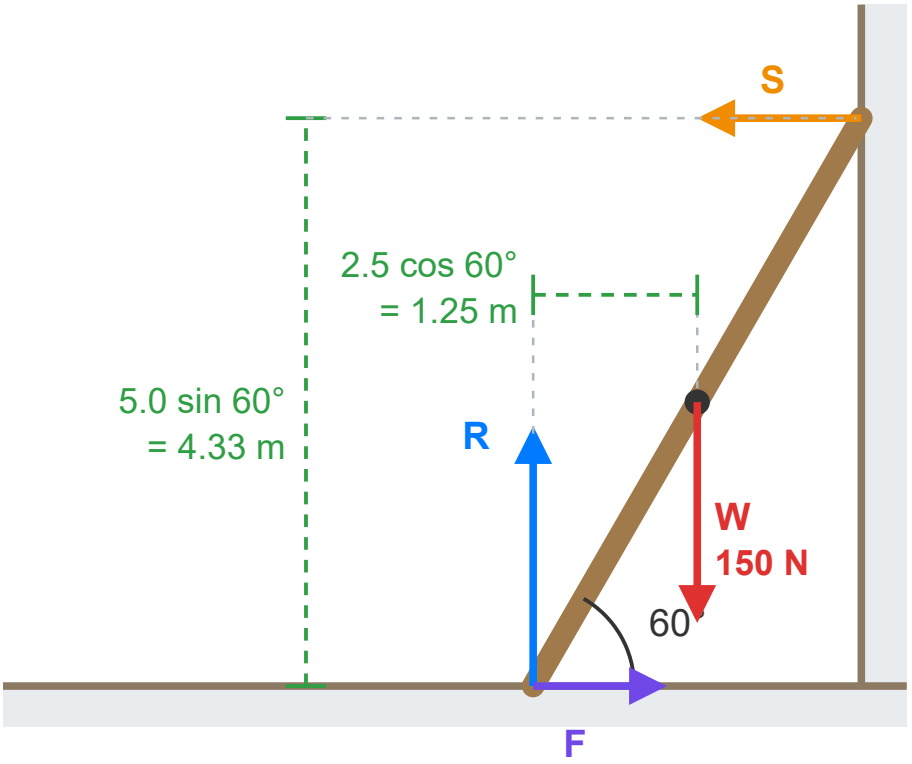
Show solution

Just as it is about to tip about B, the plank lifts off A, so $R_A = 0$. Take moments about B. The plank's weight acts 1.0 m on the A side of B; the person stands a distance $x - 3.0$ beyond B:

$$600\,(x - 3.0) = 200 \times 1.0 \quad \Rightarrow \quad x - 3.0 = 0.33 \text{ m}$$

The person can walk to 3.33 m from A, only a third of a metre past B. Try it in the simulation.

A ladder against a wall



A uniform 5.0 m ladder of weight 150 N leans on a smooth wall at 60° to the ground. The green dashed lines are the perpendicular distances from the foot of the ladder to the lines of action of S and W.

WORKED EXAMPLE

For the ladder in the diagram, calculate (a) the force S from the wall and (b) the normal reaction R and friction F at the ground.

Show solution

(a) Take moments about the **foot**, so that R and F (both acting there) drop out. S is horizontal, so its perpendicular distance from the foot is the height of the top of the ladder, $5.0 \sin 60°$. W is vertical, so its perpendicular distance is the horizontal distance to the midpoint, $2.5 \cos 60°$:

$$S \times 5.0 \sin 60° = 150 \times 2.5 \cos 60°$$

$$S \times 4.33 = 187.5 \quad \Rightarrow \quad S = 43.3 \text{ N}$$

(About the foot, W turns the ladder clockwise, towards the wall, and S turns it anticlockwise.)

(b) Resolve vertically: the only vertical forces are R and W, so $R = 150\ \text{N}$. Resolve horizontally: F must balance S, so $F = 43.3\ \text{N}$ towards the wall.

If the floor can't provide 43.3 N of friction, the foot slides out. Repeat the moments step for a general angle θ and you get $F = S = \dfrac{W}{2\tan\theta}$: the steeper the ladder, the less friction it needs, which is why ladders are safer at steep angles.

Toppling, couples and rotation

The ideas in this section go beyond the specification, but each one comes straight out of moments and explains something you see every day.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Stability and toppling

Tilt a box and let go. It falls back if the vertical line through its centre of gravity still passes **inside** its base, because the weight then has a moment about the lower edge that rotates it back. Once that line passes **outside** the base, the moment of the weight about the edge tips it over. Objects with a low centre of gravity and a wide base can tilt further before they topple, which is why racing cars are low and wide, and why a high-sided van or a lorry with a high load can tip on a sharp bend or a steep camber.

The tipping angle of a block NOT PART OF THE EDEXCEL SPEC

1. A uniform block of base width b and height h sits on a slope at angle θ. Friction stops it sliding. Its centre of gravity is $b/2$ along the slope and $h/2$ up from the base, measured from the lower corner.

2. It is on the point of toppling when the centre of gravity is directly above that corner. The line from the corner to the centre of gravity is then vertical, so it makes the angle θ with the normal to the slope:

$$\tan\theta = \frac{b/2}{h/2} \quad \Rightarrow \quad \boxed{\tan\theta_{\text{tip}} = \frac{b}{h}}$$

A cube tips at 45°. A box twice as tall as it is wide tips at only 26.6°.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Couples and torque

Turn a steering wheel with both hands and you push up on one side and down on the other. Two equal, opposite, parallel forces that don't act along the same line are called a **couple**. The resultant force is zero, so the wheel doesn't move off anywhere, but the resultant moment isn't zero, so it turns. In engineering, the turning effect of a couple is usually called a **torque**.

The moment of a couple is the same about every point

NOT PART OF THE EDEXCEL SPEC

1. Two forces F, in opposite directions, have parallel lines of action a perpendicular distance d apart. Pick any point P a perpendicular distance a from the nearer line of action, on the outside.

2. The forces turn in opposite senses about P, so their moments subtract:

$$F(a+d) - Fa = Fd$$

The a cancels: the moment of a couple is Fd whichever point you choose. That's why a couple turns an object about its centre without pushing it sideways.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Why some things are harder to spin

A resultant force gives linear acceleration, and mass resists it. A resultant moment gives **rotational** acceleration, and what resists that is the **moment of inertia**, which depends on the mass and on how far it is from the axis (each piece contributes its mass times its distance squared). A tightrope walker's long pole puts mass far from the axis, so a small unbalanced moment only tips her over slowly, giving her time to correct. A spinning ice skater who pulls her arms in moves mass closer to the axis and spins faster.

Quick checks

QUICK CHECK

The same 50 N force is applied to the end of a spanner in four different directions. Which gives the largest moment about the nut?

- Along the spanner, towards the nut
- At 30° to the spanner
- At 45° to the spanner
- At 90° to the spanner

The moment is $Fd\sin\theta$, largest when $\sin\theta = 1$. Pushing along the spanner sends the line of action through the nut, so the moment is zero.

QUICK CHECK

A uniform metre rule is pivoted at its 30 cm mark. It balances horizontally when a 0.20 kg mass hangs from the 10 cm mark. What is the mass of the rule?

- 0.067 kg
- 0.10 kg
- 0.20 kg
- 0.40 kg

The rule's weight acts at its centre, the 50 cm mark, 20 cm from the pivot. The mass is also 20 cm from the pivot on the other side, so the masses must be equal: $0.20g \times 0.20 = Mg \times 0.20$.

QUICK CHECK

Which statement must be true for an extended body in equilibrium?

- All the forces act through its centre of gravity.
- The resultant force is zero, but there can be a resultant moment.
- The resultant force is zero and the resultant moment about any point is zero.
- The clockwise moments are zero.

Equilibrium needs both conditions: no resultant force (no linear acceleration) and no resultant moment (no rotational acceleration). A couple is the classic case with zero resultant force but a non-zero moment, and it is not in equilibrium.

QUICK CHECK

A person walks along a plank that rests on two supports, from support A towards support B. What happens to the support forces?

- Both stay the same.
- The force at A decreases and the force at B increases; their sum stays the same.
- Both increase.
- The force at A increases and the force at B decreases.

Taking moments about A, R_B grows in proportion to the person's distance from A. The total weight doesn't change, so $R_A + R_B$ is constant and R_A must fall by the same amount.

QUICK CHECK

A card hung from a pin comes to rest with its centre of gravity directly below the pin. Why?

- Because the weight of the card acts at the pin.
- Only in that position does the weight have zero moment about the pin.
- Because the centre of gravity is always at the geometric centre.
- Because the pin pushes upwards harder in that position.

Anywhere else, the line of action of the weight misses the pin, so the weight has a moment about the pin that swings the card. It can only stay at rest when that moment is zero.

QUICK CHECK

A 20 N force acts on a door 0.80 m from the hinge. Its line of action is at 30° to the door. What is its moment about the hinge?

- 8.0 N m
- 13.9 N m
- 16.0 N m
- zero

The perpendicular distance is $0.80 \sin 30^\circ = 0.40$ m, so the moment is $20 \times 0.40 = 8.0$ N m. Using cos gives 13.9 N m, and ignoring the angle gives 16.0 N m.